

Expanding balloon

The radius r cm of a spherical balloon is increasing at 0.3cm/s .
Find the rate at which the volume is increasing to 3dp when
 $r=8\text{cm}$.

$$V = \frac{4}{3}\pi r^3$$

Conical pile

A conical pile of sand always has height 3 times its radius. The radius is increasing at 0.2m / min. Find the rate at which its volume is increasing when $r = 5\text{m}$.

$$V = \frac{1}{3}\pi r^2 h$$

Changing rectangle

The length of a rectangular display screen is increasing at 2 cm/s, while its width is decreasing at 0.5 cm/s.

Find the rate of change of its area when the length is 20 cm and the width is 12 cm. State whether the area is increasing or decreasing.

14cm²

Inverted Conical Tank

An inverted conical tank has height 10 m and top radius 4 m. Water drains from the bottom of the tank at a constant rate of 2 m³/min. Find the rate at which the water depth is decreasing when h=5 m.

$$V = \frac{1}{3}\pi r^2 h$$

Triangular Water Tank

A water tank is 8 m long. Each end is an isosceles triangle of height 2 m and top width 4 m. Water is pumped into the trough at 0.6 m³ /min. Find the rate at which the water depth is increasing when the depth is 1.5 m.

Surface area of right-angle triangle with two equal lengths x : $\frac{1}{2}x^2$

0.025

Fixed volume open box

A open box (no top) with a square base must hold 500cm^3 .
What is the surface area of the square base to minimize the
amount of material required to make it?

Poster Design

A poster must contain 600 cm^2 of printed material. It has margins of 3 cm on the left and right and 4 cm at the top and bottom. Find the minimal surface area of the poster.

987.396

Water Flowing Into Tank

Water enters a tank at a rate of $4t+6$ liters per minute, where t is the number of minutes after filling begins. Initially, the tank contained 20 liters of water. Find the volume of water after 5 minutes.

Distance travelled

The velocity of a car at time t is given by

$$v(t) = 3t^2 - 12t + 9$$

What's the total distance travelled by the car after 5 seconds?

$$4+4+20=28$$

Daily profit

The daily revenue rate, in dollars per day, of a company at time t days is modeled as:

$$R(t) = 8t - t^2$$

The daily cost rate is modeled as:

$$C(t) = 2t + 3$$

What is the total profit of the company over the first five days, assuming the functions are continuous?